

Jonathan Ung

Canadian Citizen | US J-1 Eligible
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EDUCATION

Simon Fraser University - Bachelor of Science in Computer Science

EXPECTED APR 2028

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Distributed Systems, Artificial Intelligence, Networking, Computer Graphics, Computer Vision, Data Science, Robotic Autonomy

EXPERIENCE

Qualcomm | Markham, ON, Canada

MAY 2026 – AUG 2026

AI Software Engineering Intern

C/C++, Python, ONNX, GenAI, QNN, Hexagon

- Unified **50+ math ops and approximation functions** across multiple repositories into a canonical header library, eliminating divergent implementations of core operations (e.g., square root approximations) and enforcing a single computation path
- Designed **MRU-based graph eviction strategy** for on-device GenAI inference, enabling models exceeding device RAM by intelligently paging KV cache graph partitions to minimize latency under memory pressure
- Built automated graph validation tool for the Hexagon Tensor Processor pipeline, detecting **8+ categories of ONNX-to-QNN compilation defects** including quantization mismatches, redundant data movers, and broken connectivity

Rivian and Volkswagen Group Technologies | Vancouver, BC, Canada

MAY 2025 – DEC 2025

Infotainment Software Engineering Intern

Kotlin, Java, Android, Protobuf, QNX

- Owned **12+ user-facing features** across Rivian R2 Media App (browsing, search, playback, shuffle, podcasts), delivering an audio visualizer and vendor integrations for Spotify, Apple Music, and Tidal to pre-production builds
- Migrated Media App from fragment-based architecture to MVVM with view controllers consuming data models, cutting RAM usage **40%** and accelerating feature velocity **30%**
- Resolved critical vehicle safety defects including cold-boot launch crashes, unsafe service data handling, and a radio refactor regression that escaped QA, collaborating across platform and QA teams
- Fixed Bluetooth AVRCP metadata errors via caching and content-hash validation, raising accuracy **from 66% to 99%**

SFU PADComp Lab | Burnaby, BC, Canada

FEB 2026 – PRESENT

Undergraduate Researcher

Go, PostgreSQL, Next.js, Electron, Docker, AWS EC2

- Developed **zone**, a per-repo Docker workspace manager for LLM coding agents with network sandboxing, adopted by **10+ researchers** for safe autonomous code execution
- Created Electron-based orchestration dashboard with real PTY terminals via **node-pty/xterm.js**, enabling researchers to monitor and debug concurrent LLM agent sessions in real time
- Built local-first observability platform for AI coding agents with **9-stage ingest pipeline**, cross-repo GitHub OAuth (**CSRF**, credential lifecycle), and a SQLite-to-PostgreSQL sync daemon with anonymization and transcript redaction on **AWS EC2**

SFU Robot Soccer | Burnaby, BC, Canada

FEB 2024 – PRESENT

Software Team Lead

C++, Python, Qt6, Protobuf, Docker

- Led **30-member C++ team**; redesigned onboarding workflows reducing new developer ramp-up time **40%**
- Engineered Behavior Tree DSL reducing node boilerplate **10:1** and Play Graph state machine with 12-play transitions evaluating **50+ metrics/tick**, coordinating 6 robots at **180Hz**
- Designed Protobuf networking protocol routing commands to **6 robots** with hot-swappable simulator/hardware modes

PROJECTS

VeTool - FPS Matchmaking Service

.NET, PostgreSQL, Next.js, SignalR, Redis, Docker

- Built full-stack matchmaking platform for FPS custom lobbies with real-time drafting, pick/ban, and match history tracking, serving **50+ active users**
- Delivered SignalR and Redis event layer achieving **sub-150ms** state sync across horizontally scaled API instances
- Designed high-concurrency PostgreSQL schema with transactional safeguards and indexed lookups for conflict-free writes

Finesse - Agentic Meta-Building Platform for Code Workflows

Python, Docker, Shell, tmux, Claude Code Plugin API

- Built meta-planning phase for Claude Code orchestrating **6 parallel validation agents** across 6 task-specific workflows
- Designed **wave-based task decomposition** with dependency-aware scheduling, automatically splitting large tasks into parallel sub-workflows across isolated git worktrees with tmux orchestration and merge reconciliation

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, Kotlin, JavaScript, TypeScript, Go, SQL, Bash

Frameworks: React, Next.js, .NET, FastAPI, Node.js, Express.js, Spring Boot, Electron, SignalR, GraphQL

Tools & Infrastructure: Docker, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB, Redis, Git, GitHub & GitLab CI/CD, Linux

Concepts: API Design, Distributed Systems, Developer Tooling, Real-Time Systems, LLMs, Applied Machine Learning, Robotics